`DiffAlg`: A DIFFERENTIAL ALGEBRA PACKAGE

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Abstract. In this article we present `DiffAlg`, a differential algebra package for Macaulay2. It can perform the following operations: wedge products and exterior differential of differential forms, contraction and Lie derivative of a differential form with respect to a vector field and Lie brackets between vector fields.

Given a homogeneous differential operator of degree one D, the lack of an algebraic module structure attached to the kernel or image of D hinders the study of D. The main purpose of `DiffAlg` is to handle these spaces degree-wise.

1. Motivation and description of the package

Algebraic and differential operations arise naturally when working with differential forms and vector fields, *e.g.*: wedge products and exterior differential of differential forms, contraction and Lie derivative of a differential form with respect to a vector field and Lie brackets between vector fields. Some important statements involving these operations include the following:

(a) A differential r-form ω in the affine space $\mathbb{K}^{n+1}$ *descends to the projective space* $\mathbb{P}^n_{\mathbb{K}}$, if it satisfies the equation
$$i_R\omega = 0,$$
where $\mathbb{K}$ is a field, R is the radial vector field $R = \sum x_i \frac{\partial}{\partial x_i}$ and i_R denotes the contraction, see [3, Theorem 8.13, p. 176].

(b) If ω is a differential 1-form, then ω defines a foliation in $\mathbb{K}^{n+1}$ if it satisfies the *Frobenius integrability condition* given by the equation
$$\omega \wedge d\omega = 0,$$
see [4, Definition 2.2, p. 823].

(c) Let ω be an integrable 1-form and $L_X\omega$ the Lie derivative of ω with respect to a vector field X. Then, the solutions of the equation
$$L_X\omega = 0$$
define all the *infinitesimal automorphisms* of the foliation given by ω, see [4, Proposition 7.7, p. 845].

(d) Let ω be an integrable 1-form. Then the *tangent space of the space of foliations* at ω is given by the differential 1-forms η that satisfy the equation
$$\omega \wedge d\eta + d\omega \wedge \eta = 0,$$

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see [2, Section 2.1. p. 709].

(e) Let D be a *bracket generating distribution*. Some important invariants of D are the ranks of the derived sequence

$$a(p) := \operatorname{rank} D^{(p)} = \operatorname{rank}\left(D^{(p-1)} + [D, D^{(p-1)}]\right),$$

see [5, §1, pp. 8-9].

(f) A *symplectic structure* in a variety of dimension $2r$ is given by a 2-form ω such that $d\omega = 0$ and $\omega^r \neq 0$, see [1, p. 41].

For a clear understanding on how `DiffAlg` deals with such equations, let us fix some notation.

Let $S = \mathbb{K}[x_0, \ldots, x_n]$ be the polynomial ring in $n+1$ variables and let $\Omega = \bigoplus_{r\geq 0} \Omega^r$ be the exterior algebra of differential forms of S over $\mathbb{K}$. Let $\Omega^r(d)$ denote the space of r-forms with polynomial coefficients of homogeneous degree d, where we assign degree 1 to each x_i and degree 0 to each dx_i.

Therefore $\omega \in \Omega^r(d)$ can be written as

$$\omega = \sum_{\substack{I\subset\{0,\ldots,n\}\\ \#I=r}} \sum_{\substack{\alpha\in\mathbb{N}^{n+1}\\ |\alpha|=d}} a_{\alpha,I}\ x^\alpha\ dx_I, \qquad a_{\alpha,I} \in \mathbb{K}, \tag{1}$$

where for each I of the form $I = \{i_1, \ldots, i_r\} \subset \{0, \ldots, n\}$ let dx_I denote $dx_{i_1} \wedge \ldots \wedge dx_{i_r}$ and each $\alpha = (\alpha_0, \ldots, \alpha_n) \in \mathbb{N}^{n+1}$ denote $|\alpha| := \sum_{i=0}^n \alpha_i$ and $x^\alpha := x_0^{\alpha_0} \cdot \ldots \cdot x_n^{\alpha_n}$.

Let T be the module of vector fields with coefficients in S. Let $T(e)$ denote the homogeneous vector fields with polynomial coefficients of degree e, where we assign degree 1 to each x_i and degree 0 to each $\frac{\partial}{\partial x_i}$ and, analogously to eq. (1), $X \in T(e)$ can be written as

$$X = \sum_{i=0}^{n} \sum_{\substack{\beta\in\mathbb{N}^{n+1}\\ |\beta|=e}} b_{\beta,i}\ x^\beta\ \frac{\partial}{\partial x_i}, \qquad b_{\beta,i} \in \mathbb{K}.$$

Current algebraic software systems implement functionality to deal with differential forms and vector fields, but usually scalar parameters $a_{\alpha,I}$ and $b_{\beta,i}$ must be specified as fixed elements in $\mathbb{K}$. Instead, `DiffAlg` treats homogeneous forms and vector fields in a completely symbolic environment by considering the scalar coefficient rings

$$\mathbb{K}[a_{\alpha,I}] \qquad \text{and} \qquad \mathbb{K}[b_{\beta,i}].$$

Scalar coefficients can be systematically obtained by looking at the coordinates of differential forms and vector fields written in the standard basis

$$\mathcal{B}_{r,d} = \{x^\alpha dx_I\}_{\substack{\#I=r\\ |\alpha|=d}} \qquad \text{and} \qquad \mathcal{B}_e = \left\{x^\beta \tfrac{\partial}{\partial x_i}\right\}_{|\beta|=e}$$

of the spaces $\Omega^r(d)$ and $T(e)$, respectively.

Importantly, when using `DiffAlg`, each object is expected to be defined in its own coefficient ring. Then, computing some operations (wedge product, contraction, etc.) will involve different input and output rings, producing a modification of the coefficients rings $\mathbb{K}[a_{\alpha,I}]$ or $\mathbb{K}[b_{\beta,i}]$. For greater clarity, consider the following

example. Fix $\omega \in \Omega^r(d)$ and $X \in T(e)$ and consider the contraction $i_X\omega \in \Omega^{r-1}(d+e)$. Then, the following elements will be taking place:

Operation	(ω, X)	$\longmapsto$	$i_X\omega$
Rings	$\mathbb{K}[a_{\alpha,I}] \times \mathbb{K}[b_{\beta,i}]$		$\mathbb{K}[a_{\alpha,I}, b_{\beta,i}]$
Basis	$\mathcal{B}_{r,d} \times \mathcal{B}_e$		$\mathcal{B}_{r-1,d+e}$

As mentioned before, the main purpose of `DiffAlg` is to find algebraic solutions to equations in the context of differential algebra. Equations are treated differently in the linear and non-linear cases:

i) In the linear case, for example $i_R\omega = 0$, `DiffAlg` can compute a basis of the solutions of the equation. Once this is done, it can also compute a generic linear combination of the elements of the basis, see Example 1.
ii) In the non-linear case, for example $\omega \wedge d\omega = 0$, the coordinates will be polynomial. In this case `DiffAlg` would compute the ideal generating the space of solutions. This ideal can be obtained in two different ways: taking coordinates in the basis $\mathcal{B}_{r,d}$ or $\mathcal{B}_e$ or taking coordinates in the basis $\{dx_I\}$ or $\{\frac{\partial}{\partial x_i}\}$, see Examples 2 and 4.

`DiffAlg` can also be a valuable tool for studying differential operators. The lack of an algebraic theory to deal with such objects can be mitigated by non-conclusive computations easily made by `DiffAlg`. As a first example, one could consider computing solutions of the annihilation of a differential operator degree-wise for low degrees.

2. Some examples

Example 1. In the following example we obtain a basis of the space of projective differential 2-forms in $\mathbb{P}^3_{\mathbb{K}}$. Then, we define a generic projective differential 2-form to be possibly used in further computations.

```
i1 : loadPackage "DiffAlg";

i2 : R = radial 3

o2 = x ax  + x ax  + x ax  + x ax
      0  0    1  1    2  2    3  3

o2 : DiffAlgField
```

This creates radial vector field in 4 variables. We are denoting the basic field $\frac{\partial}{\partial x_i}$ as ax_i .

```
i3 : w = newForm(3,2,1,"a");

o3=(a x +a x +a  x +a  x )dx dx +(a x  + a x  + a  x  + a  x )dx dx
     0 0  6 1  12 2  18 3   0  1   1 0    7 1    13 2    19 3   0  2
     ----------------------------------------------------------------
     + (a x  + a x  + a  x  + a  x )dx dx  + (a x  + a x  + a  x  +
         3 0    9 1    15 2    21 3   1  2     2 0    8 1    14 2
     ----------------------------------------------------------------
     a  x )dx dx  + (a x  + a  x  + a  x  + a  x )dx dx +(a x +a  x  +
      20 3   0  3     4 0    10 1    16 2    22 3   1  3   5 0  11 1
     ----------------------------------------------------------------
     a  x  + a  x )dx dx
      17 2    23 3   2  3

o3 : DiffAlgForm
```

This creates a generic linear 2-form, in $\Omega^2_{\mathbb{K}^4}(1)$, where the coefficients are indexed as a_i.

Note that the ring where w is defined is

```
i4 : pretty ring w

      QQ[i]
o4 = ------[][a , a , a , a , a , a , a , a , a , a , a , a ,
      2        0   1   2   3   4   5   6   7   8   9   10  11  12
     i  + 1
     ----------------------------------------------------------------
      a  , a  , a  , a  , a  , a  , a  , a  , a  , a  , a  ][x , x ,
       13   14   15   16   17   18   19   20   21   22   23   0   1
     ----------------------------------------------------------------
     x , x ][dx , dx , dx , dx ]
      2   3    0    1    2    3
```

```
i5 : K = genKer (R _ w, w);

With this we get a basis (as a Macaulay2 list) of the forms that
descend to projective space. The operation R _ w computes the
contraction of the differntial form w with the vector field R.

i6 : length K

This way we get the dimension of Ω^2_{P^3}(1) in projective 3-space

o6 = 4

i7 : v = linearComb(K,"a")

Finally, with this last operation we define a generic projective form
where, again, we are chossing as a_i as the coefficients.

o7 = (a x  - a x )dx dx +(- a x  + a x )dx dx  + (a x  + a x )dx dx +
       0 2    1 3   0  1     0 1    2 3   0  2     0 0    3 3   1  2
     ----------------------------------------------------------------
     (a x  - a x )dx dx  + (-a x  - a x )dx dx  + (a x  + a x )dx dx
       1 1    2 2   0  3      1 0    3 2   1  3     2 0    3 1   2  3

o7 : DiffAlgForm

Note that the working ring, attached to the element v, has now
changed to:

i8 : pretty ring v

       QQ[i]
o8 = ------[][a , a , a , a ][x , x , x , x ][dx , dx , dx , dx ]
       2        0   1   2   3   0   1   2   3    0    1    2    3
      i  + 1
```

Example 2. In the finite dimensional $\mathbb{K}$-vector space $\Omega^1(d)$, the solutions of the equation $\omega \wedge d\omega = 0$ determine an algebraic variety, its points are the integrable differential 1-forms of degree d. In the following example, we compute the equations of the variety of integrable 1-forms of degree 1 in 3-dimensional space.

It is worth mention that, for $n \geq 3$ and $d > 5$, it is an open problem to classify the irreducible components of this varieties, see [2].

```
i1 : loadPackage "DiffAlg";

i2 : w = newForm (2,1,1,"a")

This creates a generic linear 1-form in Ω^1_{K^3}(1), with coefficients a_i.

o2 = (a x  + a x  + a x )dx  + (a x  + a x  + a x )dx +(a x  + a x +
       0 0    3 1    6 2   0     1 0    4 1    7 2   1   2 0    5 1
```

```
      ----------------------------------------------------------------
        a x )dx
         8 2   2

o2 : DiffAlgForm

i3 : moduliIdeal (w ^ (diff w))

This operation returns the ideal of the scalar coefficients given by
w ^ (d w) = 0.

o3 = ideal (- a a  + a a  + a a  - a a , - a a  + a a  + a a  - a a ,
                2 3    0 5    1 6    0 7     2 4    1 5    4 6    3 7
     ----------------------------------------------------------------
            a a  - a a  + a a  - a a )
             5 6    2 7    1 8    3 8

                QQ[i]
o3 : Ideal of ------[][a , a , a , a , a , a , a , a , a ]
                2        0   1   2   3   4   5   6   7   8
               i  + 1
```

Example 3. Let D be a 2-dimensional distribution generated by vector fields X and Y in 5-dimensional space. In the following example we compute the ranks of the derived distributions $D^{(p)}$. We verify that D is bracket-generating, *i.e.*, its derived series eventually spans the entire tangent space.

```
i2 : X = newField "x_0^2*ax_0+ x_1^2*ax_1+ x_2^2*ax_2+ x_3^2*ax_3" ;

i3 : Y = newField "x_5*ax_0+ x_4*ax_1+ x_3*ax_2+ x_2*ax_3+ x_1*ax_4+
      x_0*ax_5";

i4 : D_0 = {X,Y};

i5 : for b in 1..3 do
     (for a in D_(b-1) do
     (D_b=join(D_(b-1),{a|Y,a|X})));

With this we are computing the derived sequence

i6 : {rank dist D_0, rank dist D_1, rank dist D_2, rank dist D_3}

o6 = {2, 3, 5, 6}

This are the ranks of the derived series

o6 : List
```

Example 4. In the following example, we generate a random rational 1-form of type (1,2) in $\mathbb{P}^2_{\mathbb{K}}$. First, we compute (the dimension of) the space of its integrating factors, see [4, pp. 828-829]. Then, we compute the ideal of the singular locus of ω, *e.g.*, the ideal where ω vanishes.

```
i2 : w = random logarithmicForm (2,{1,2},"a",Projective => true);

i3 : f = newForm (2,0,3,"a");

i4 : length genKer(w^(diff f)+f * (diff w),f)

o4 = 2

i5 : I = singularIdeal w

                          2                          2    2
o6 = ideal (- 9x x  + 63x  - 36x x  - 54x x  - 54x , 9x  - 63x x  -
                 0 1      1      0 2      1 2      2    0      0 1
     ----------------------------------------------------------------
                   2     2                2
      27x x  -45x x  - 54x , 36x  + 81x x  + 45x  + 54x x  + 54x x )
         0 2     1 2      2     0      0 1      1      0 2      1 2

                QQ[i]
o6 : Ideal of ------[][x , x , x ]
                2        0   1   2
               i  + 1
```

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